# Zhidan Li

Shanghai Jiao Tong University, Minhang, Shanghai, China
Email: lizhidan17@gmail.com
https://lizhidan00.github.io
Latest Update: October 20, 2026

**Experience**

- *Visiting Postgraduate Research Student* 2022.08 - 2022.10
  School of Informatics, University of Edinburgh
  Supervisor: *Heng Guo*

**Education**

- *Doctor of Philosophy* in Computer Science 2023.09 - current
  John Hopcroft Center, Shanghai Jiao Tong University
  Advisor: *Kuan Yang*
- *Bachelor of Engineering* in Computer Science 2019.09 - 2023.06
  ACM Honor Class, Shanghai Jiao Tong University

**Research Interests**

- Approximate counting and sampling in theoretical computer science.
- Counting and sampling problems on random structures.
- Algorithmic implications from statistical physics.
- High-dimensional probability and geometry.

**Research Articles**

- Rapid Mixing for Sampling Colorings of Random Graphs up to Tree Uniqueness Threshold
  **Zhidan Li** and Kuan Yang
  Preprint.
- Fast Almost-Uniform Sampling of Random $k$-SAT Solutions
  Kun He, **Zhidan Li** and Kuan Yang
  Preprint.
- Spatial mixing and deterministic approximate counting of multi-spin systems beyond bounded degree graphs
  **Zhidan Li** and Kuan Yang
  Preprint.
- An FPRAS for antiferromagnetic Ising models on random regular bipartite graphs
  **Zhidan Li** and Kuan Yang
  Preprint.
- Counting and sampling anti-ferromagnetic Potts models on random regular bipartite graphs in the non-uniqueness regime
  **Zhidan Li**, Siyu Liu and Kuan Yang
  Preprint.
- FPTAS for Holant problems with log-concave signatures
  Kun He, **Zhidan Li**, Guoliang Qiu and Chihao Zhang
  In *Proceedings of the 2025 Annual ACM-SIAM Symposium on Discrete Algorithms* (**SODA 2025**).

**Teaching**

- In Shanghai Jiao Tong University:
  - Serve as a *Teaching Assistant*:
    - CS3905: *Algorithm Design and Analysis* Spring 2023
      Instructor: *Yuhao Zhang*
    - CS2910: *Optimization* Summer 2023
      Speaker: *Yunwei Ren*
    - MATH3806: *Optimization Methods* Autumn 2023
      Instructors: *Bo Jiang* and *Kuan Yang*
    - CS0901: *Combinatorics* Autumn 2024, 2025, 2026
      Instructor: *Kuan Yang*

**Seminars**

- In Shanghai Jiao Tong University:
  - Serve as a *Speaker*:
    - Reading Seminar on *Localization Schemes* Spring 2023
      Organizer: *Chihao Zhang*
    - Reading Seminar on *Introductions to Random Graphs* Spring 2024
      Organizer: *Kuan Yang*